

Ziheng He 何子恒

Ph.D. Student in Artificial Intelligence

University of Chinese Academy of Sciences · Institute of Automation, CAS

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Embodied world models, video generation and diffusion, and robot learning from generated data.



Education

**University of Chinese Academy of Sciences**2026 - Present

Ph.D. in Artificial Intelligence · School of Advanced Interdisciplinary Sciences (SAIS)
Research affiliation: Institute of Automation, Chinese Academy of Sciences · NLPR

**Southeast University | B.Eng. in Software Engineering**2022.09 - 2026.06

Rank: 9/86 · Outstanding Graduate · Merit Student · Outstanding Undergraduate Thesis

Publications & Preprints

SKIP: Sparse Keyframe Interpolation Paradigm for Efficient Embodied World Models

Ziheng He, Yixiang Chen, Ning Yang, et al.

Conference on Robot Learning (CoRL) 2026 · Accepted · First author · arxiv.org/abs/2606.00664

Event-preserving sparse-to-dense world modeling; 4.16x faster rollout and 89.0% lower FVD on LIBERO versus a dense baseline.

GigaBrain-0.7: Scaling Embodied Foundation Models to Emergent Capabilities with a Three-System Architecture

GigaBrain Team (including Ziheng He)

arXiv preprint, 2026 · Co-author · arxiv.org/abs/2608.15875

XEWorld: Can Action-Conditioned World Models Generalize to Unseen Robot Embodiments?

Yixiang Chen, Jiabing Yang, Yuan Xu, Qisen Ma, Keji He, Peiyan Li, Kai Wang, Ziheng He, et al.

arXiv preprint, 2026 · Co-author · arxiv.org/abs/2608.05799

WAM-Nav: Asymmetric Latent World-Action Modeling for Unified Visual Navigation

Ning Yang, Yan Huang, Kaiwen Peng, Ziheng He, et al.

arXiv preprint, 2026 · Co-author · arxiv.org/abs/2606.04907

Research Experience

**GigaAI | Research Intern**2025.09 - Present

- GigaBrain-0.7 co-author. Trained System 3, a 5B World Value Model, and designed its visual-subgoal and progress-conditioning interface for the action policy.

**FiveAges | Research Intern**2025.09 - Present

- First-author work on SKIP: event-preserving keyframe selection, sparse video generation, and action-conditioned interpolation for efficient embodied world models.
- Built FiveagesWAM with sparse-frame inference and historical-image memory for continuous video prediction.

**Huawei Nanjing Research Center · Celia Team | AI Algorithm Intern**2025.06 - 2025.09

- Worked on on-device LLM inference: speculative decoding, KV-cache and parallel-decoding optimization, and CUDA profiling with Nsight.

Open Source & Research Tools

**AI Bro | Creator & Maintainer**github.com/zihenghe04/AIBro

Local-first AI workspace connecting conversations, project files, source-linked knowledge, tasks, and schedules. Includes retrieval, reviewable file changes, and reusable agent workflows.
Public macOS preview · v0.7.9 released September 17, 2026

**CCDash | Creator & Maintainer**github.com/zihenghe04/CCDash

Unified usage dashboard for Claude Code, claude.ai, and API traffic: tokens, quotas, and costs.

**AionUi | Contributor**github.com/iOfficeAI/AionUi

Contributed a merged bilingual guide for headless Linux deployment, covering Xvfb, service management, and remote access.